

Behavior of Novel Beam-strengthened Embedded CFST Column Bases under Cyclic Loading: Experimental Study and Strength Prediction

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Abstract

Embedded column base connections are widely used in high-rise buildings due to their favorable seismic performance and robust load transfer to foundations. However, with the increasing adoption of large cross-section columns in engineering practice, such connections often face challenges in satisfying code-specified embedment depth requirements, particularly under constraints of limited underground space. To address the issues of inadequate energy dissipation and insufficient load-carrying capacity in shallowly embedded column bases, this study proposes a novel beam-strengthened embedded concrete-filled steel tubular (CFST) column base connection.

Cyclic lateral loading tests were conducted on two beam-strengthened specimens with relatively shallow embedment and two corresponding conventional embedded column base (ECB) specimens. Based on the experimental results, a refined finite element (FE) model was developed and validated. A comprehensive parametric study was subsequently carried out to evaluate the effects of key parameters, including embedment depth, axial load ratio, geometric dimensions, and material strength, with the aim of elucidating the load-transfer mechanism and quantifying the contribution of the strengthening beams.

Based on the experimental and numerical results, a theoretical method was proposed for predicting the ultimate strength of the novel connections. The results demonstrate that the incorporation of strengthening beams significantly alleviates pinching and sliding effects in the hysteretic response, while also reducing cracking in the concrete foundation. The proposed connection exhibits enhanced load-carrying capacity and seismic performance under shallow embedment conditions, providing an effective and practical solution for engineering applications.

Key words: Embedded column base connection; Concrete-filled steel tube; Strengthening Beam; Seismic performance; Strength prediction

1. Introduction

The column base constitutes a critical interface between the first-story column and the reinforced concrete foundation, ensuring effective transfer of loads from the upper structure to the foundation. With the rapid proliferation of high-rise construction, embedded column bases have been widely adopted in steel structures owing to their favorable seismic performance [1-3].

Nevertheless, several challenges remain in the practical application of embedded steel column base connections. A key concern lies in the minimum embedment depth requirements prescribed by current design codes to guarantee sufficient connection resistance. For instance, JGJ 138-2016 [4] specifies that the embedment depth should not be less than twice the column section depth. Similarly, JGJ 99-2015 [5] requires embedment depth-to-section depth ratios of 2.0 for H-section steel columns and 2.5 for box-section columns. Such stringent provisions may impose significant constraints on engineering practice, particularly when large-scale steel or concrete-filled steel tubular (CFST) columns are employed, thereby limiting the constructability and economic efficiency of embedded column base systems.

To address these limitations, numerous studies have been conducted to enhance the load-carrying capacity of column base connections while reducing the required embedment depth. Hsu and Lin [6] investigated the seismic performance of rectangular CFST embedded column base connections with and without additional vertical stiffeners (Fig.1a and b). Moon *et al.* [7] experimentally examined circular CFST embedded column base connections (Fig.1c) and demonstrated that parameters such as embedment depth, axial load ratio, and shear reinforcement ratio significantly influence the strength and ductility of the connections.

Tan and Guo *et al.* [8] proposed a novel CFST embedded column base incorporating shear studs to enhance punching shear resistance (Fig.1d). Push-out test results indicated that the punching shear capacity increased by more than 300% compared with specimens without studs. In addition, Tan *et al.* [9] investigated a column base connection strengthened with a pair of steel beams and anchor bolts (Fig.1e). Hysteretic test results showed that, for a specimen with an embedment depth-to-column diameter ratio of 0.7, the inclusion of stiffening steel beams increased the ultimate load-carrying capacity and energy dissipation by 12.9% and 37.2%, respectively, compared with the unstiffened counterpart.

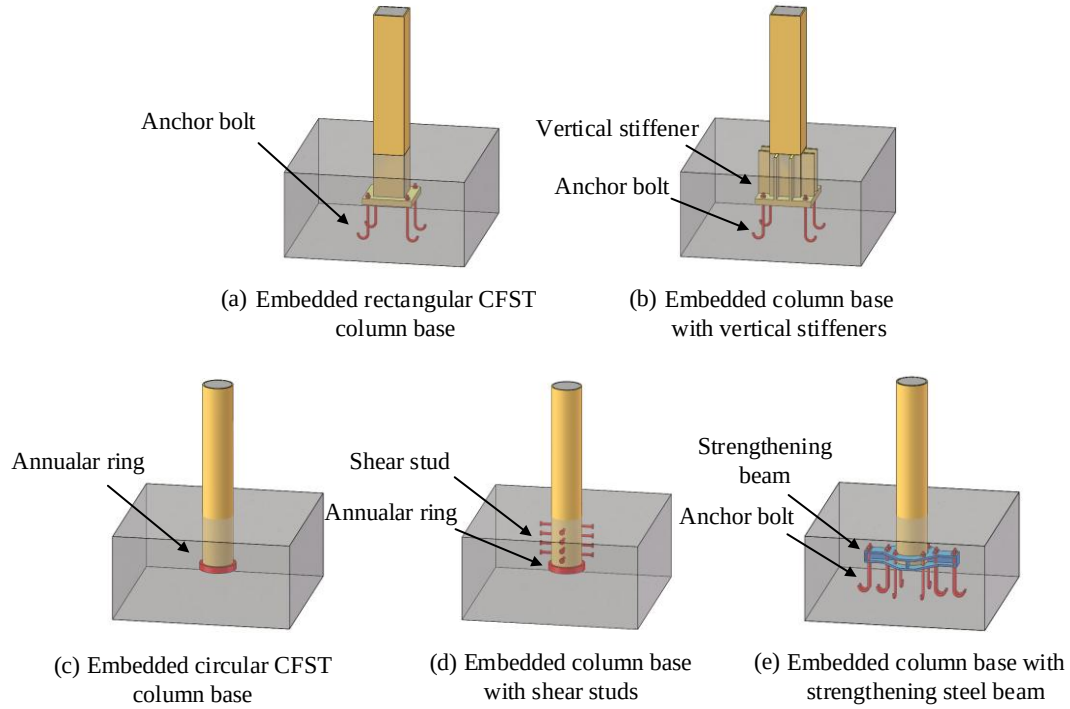


Fig.1 Existing strengthening methods for embedded CFST column bases

This study proposes a novel beam-strengthened embedded CFST column base connection and systematically investigates its load transfer mechanism and failure modes through cyclic loading tests and numerical simulations. Based on the analytical results, a theoretical model is subsequently developed to predict the ultimate strength of the proposed connection.

2. Experimental study

2.1 Test program

Two specimens of beam-strengthened embedded CFST column bases (BSECB) were designed and tested in this study (Table 1), where two reference specimens without strengthening beam (ECB) were also tested.

Table 1 Test specimens

Specimen	d_{embed} (mm)	Steel tube ($H \times B \times t$)	Strengthening beam	
			Section	L_s
			($h_s \times b_s \times t_w \times t_f$)	(mm)
BSECB300	300	250×250×10	244×175×7×11	375
ECB300	300	250×250×10	-	-
BSECB400	400	250×250×10	244×175×7×11	375
ECB400	400	250×250×10	-	-

As summarized in Table 1, all specimens employed cold-formed steel tubes for the CFST columns, with a cross-section of 250 mm × 250 mm × 10 mm ($H \times B \times t$, see Fig.2a). For the

BSECB specimens, two strengthening beams were welded to the CFST column along the loading direction, with the bottom flanges aligned with the base of the column. The strengthening beams adopted identical I-sections, with a height (h_s) of 244 mm, width (b_s) of 175 mm, web thickness (t_w) of 7 mm, and flange thickness (t_f) of 11 mm.

Diaphragms with a thickness of 10 mm were installed inside the steel tube at the levels of both beam flanges. To facilitate concrete casting, a circular opening with a diameter of 100 mm was provided in the upper diaphragm, while a smaller circular hole (20 mm in diameter) was introduced in the lower diaphragm to allow air to escape during concrete casting.

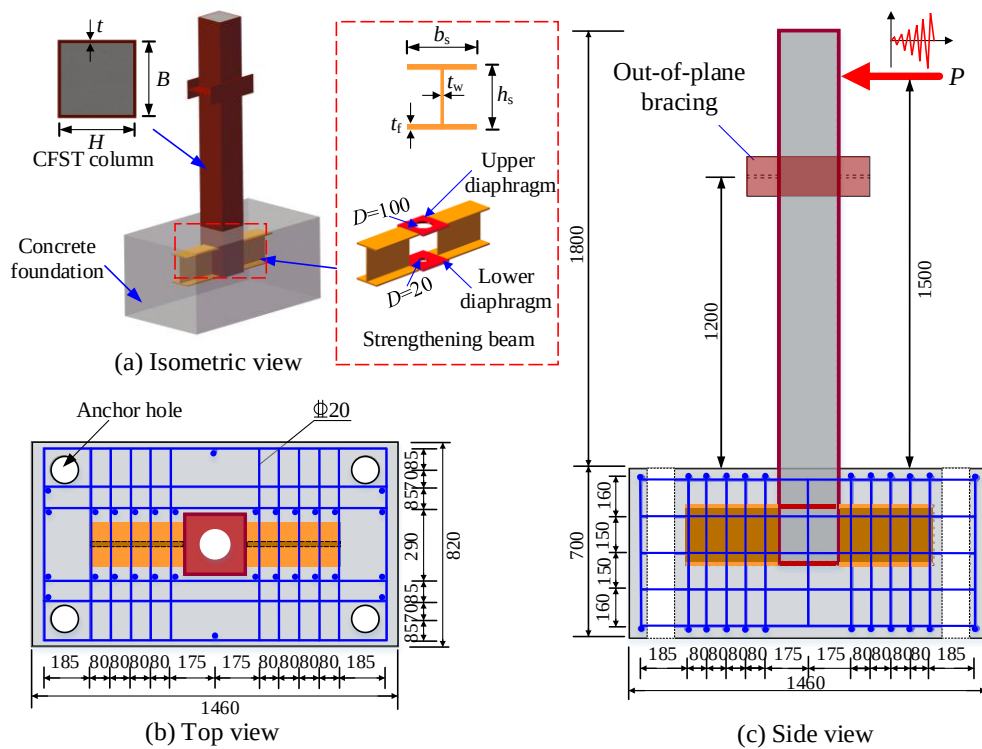


Fig.2 Details of the test specimens (mm)

For each specimen, the CFST column was embedded in a reinforced concrete block to simulate the foundation. The concrete block had dimensions of 1460 mm (length), 700 mm (height), and 820 mm (width), as shown in Fig.2b and c. The reinforcement layout is illustrated in Fig.2b and c, where ribbed steel bars with a diameter of 20 mm were used for the stirrups as well as the longitudinal and vertical reinforcements. Four anchor holes (70 mm in diameter) were positioned near the corners of the concrete block to facilitate anchorage to the strong floor using anchor bolts with a diameter of 60 mm. Cyclic lateral loading was applied at a height of 1500 mm above the top surface of the

concrete block using an MTS hydraulic actuator. To prevent out-of-plane deformation during testing, a lateral bracing system was installed at a height of 1200 mm (Fig.3). It should be noted that the reference specimens, ECB300 and ECB400, were identical to BSECB300 and BSECB400 in all respects except for the absence of strengthening beams.

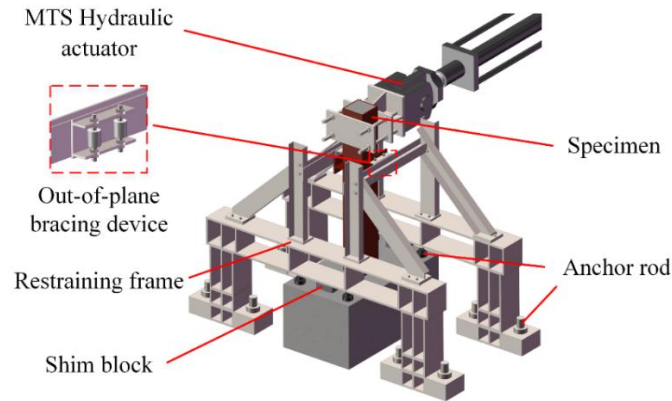


Fig.3 Test setup

2.2 Test results

2.2.1 Failure modes

As shown in Fig.4a-d, both specimens with strengthening beams (BSECB300 and BSECB400) exhibited failure characterized by local buckling of the steel tube above the top surface of the concrete block (Fig.4c and d). The onset of visible local buckling was observed at story drift ratios of approximately 3.5% - 4.5%.

In contrast, the failure modes of the reference specimens were governed by either crushing of the surrounding concrete (ECB300) or local buckling of the steel tube (ECB400), depending on the embedment depth. For specimen ECB300, pronounced interfacial separation between the CFST column and the concrete block along the loading direction was observed at a drift ratio of approximately 2.0% (Fig.4a). At this stage, severe concrete crushing occurred in front of the column, while no noticeable local buckling of the steel tube was observed throughout the loading process.

By comparison, specimen ECB400, with a greater embedment depth of 400 mm, exhibited enhanced anchorage capacity, leading to a shift in the governing failure mode to local buckling of the steel tube, which is characteristic of CFST columns.

During testing, cracking was observed in the concrete blocks of all specimens (Fig.4a-d). As shown in Fig.4a, specimen ECB300 exhibited extensive cracking in the concrete

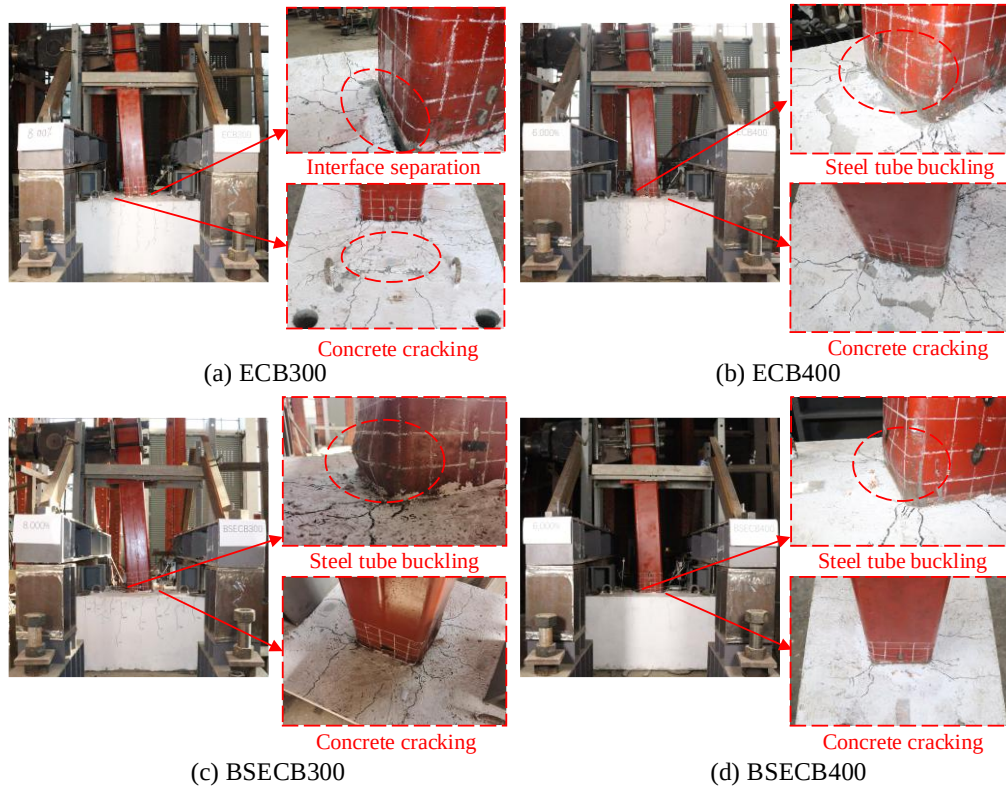


Fig.4 Failure mode of specimens

surrounding the CFST column, whereas the remaining specimens showed relatively minor cracking (Fig.4b-d). The cracks were observed to initiate near the corners of the CFST column and subsequently propagate toward the edges of the concrete block.

At failure, the cracks in specimen ECB300 had extended over nearly the entire top surface of the concrete block, except for two small regions adjacent to two sides of column. In contrast, for the other specimens (ECB400, BSECB300, and BSECB400), cracking remained confined to limited regions near the column corners. In addition, a series of vertical cracks developed on the side faces of the concrete blocks in specimens ECB300, BSECB300, and BSECB400, whereas such cracking was relatively limited in specimen BSECB400.

2.2.2 Hysteresis curves

Fig.5a-d present the hysteretic responses of all specimens, where P , M , δ and θ denote the lateral force, the bending moment at the reference section (i.e., at the top surface of the concrete block), the lateral displacement at the loading point, and the rotation angle ($\theta = \delta/1500$ mm), respectively.

It should be noted that the maximum rotation angles achieved by specimens ECB400 and BSECB400 were approximately 7%, slightly lower than those of ECB300 and BSECB300,

which exceeded 8% (Fig.5a-d). This difference arises because the tests for ECB400 and BSECB400 were terminated at a rotation angle of about 7% to improve testing efficiency, as no additional significant phenomena were observed in ECB300 and BSECB300 beyond this level of deformation.

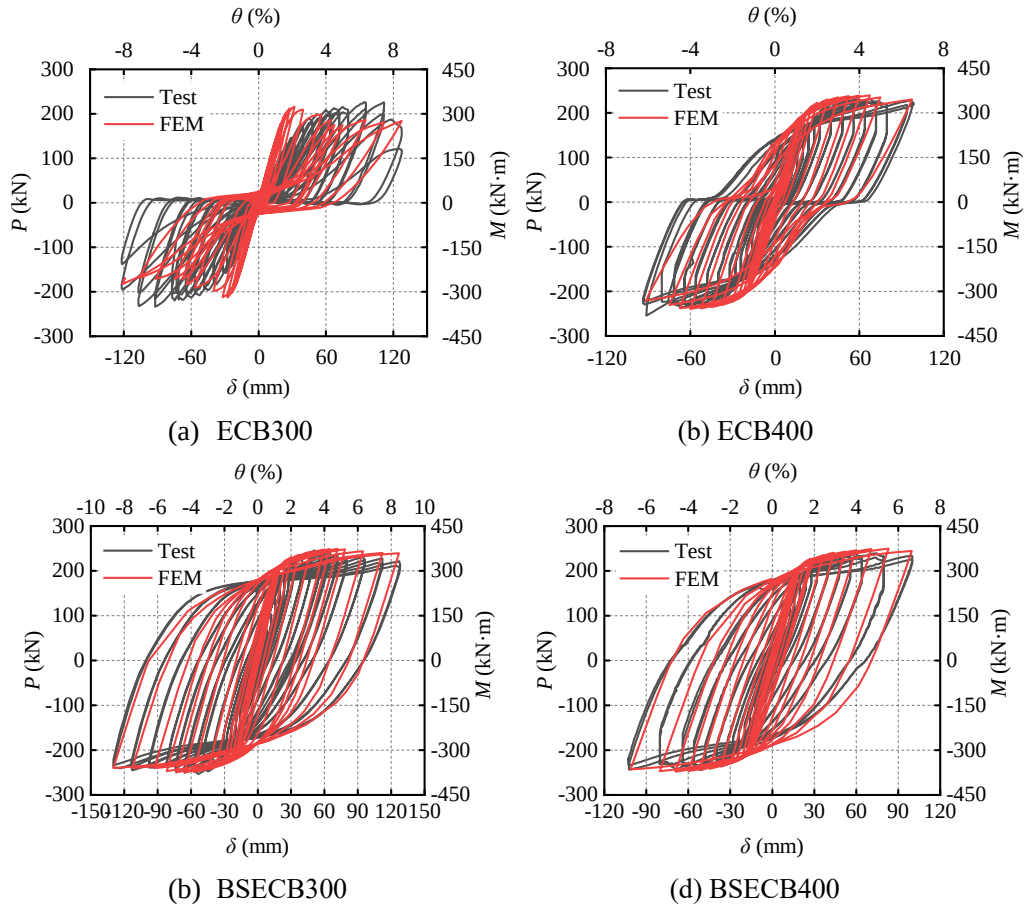


Fig.5 Hysteresis curves of specimens

The hysteretic curves of the specimens with strengthening beams (BSECB300 and BSECB400) exhibit a stable fusiform shape, with only minor pinching observed in the second and fourth quadrants (Fig.5c and d). In contrast, pronounced pinching is evident in the hysteretic responses of the reference specimens (ECB300 and ECB400, see Fig.5a and b), particularly for ECB300.

In addition to the pinching effect, noticeable sliding behavior is observed near the horizontal axis of the hysteretic loops for specimens ECB300 and ECB400, which can be attributed to concrete crushing around the CFST columns. This sliding response is especially significant in specimen ECB300 (Fig.5a), resulting in an approximately Z-shaped hysteretic response under large displacement cycles.

2.2.3 Skeleton curves and stiffness degradation

The skeleton curves, obtained by connecting the peak load points of each loading cycle from the hysteretic responses, are presented in Fig.6. It can be observed that the skeleton curves of the specimens with strengthening beams (BSECB300 and BSECB400) are nearly identical throughout the loading process. In contrast, the reference specimens (ECB300 and ECB400) exhibit lower peak loads and reduced stiffness compared with their strengthened counterparts.

For the specimen with a relatively shallow embedment depth (ECB300), pronounced reductions in both stiffness and peak load are evident relative to the other specimens. Furthermore, a rapid strength degradation is observed at a rotation angle of approximately 7%, which is likely attributable to the onset of pull-out effects on the tension side under large bending deformations.

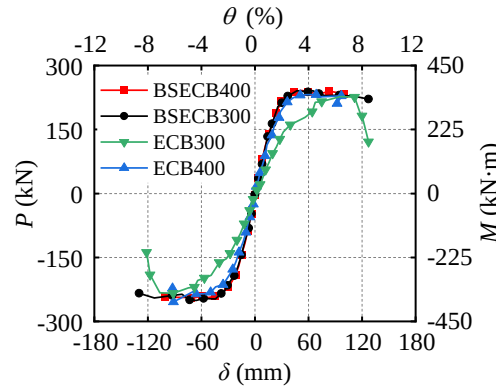


Fig.6 Skeleton curves

Fig.7 illustrates the stiffness degradation of all specimens, characterized by the stiffness degradation coefficient (K_i), which reflects the progressive accumulation of concrete damage and steel plasticity. According to [10], K_i can be given by Eq.(1):

$$K_i = \frac{|-P_i| + |+P_i|}{|-\delta_i| + |+\delta_i|} \quad (1)$$

where $+P_i$ and $-P_i$ represent the positive and negative peak loads of each loading circle, and $+\delta_i$ and $-\delta_i$ are the corresponding displacements, respectively.

As shown in Fig.7, the stiffness degradation trends with increasing lateral displacement are nearly identical for specimens BSECB300 and BSECB400. Specimen ECB400 exhibits a comparable initial stiffness to the two beam-strengthened specimens; however, its stiffness degrades at a slightly faster rate as the rotation angle increases from 0 to 4%. In contrast,

specimen ECB300 shows a significantly lower initial stiffness than the other specimens. As the rotation angle exceeds approximately 6%, the residual stiffness values of all specimens tend to converge and become comparable.

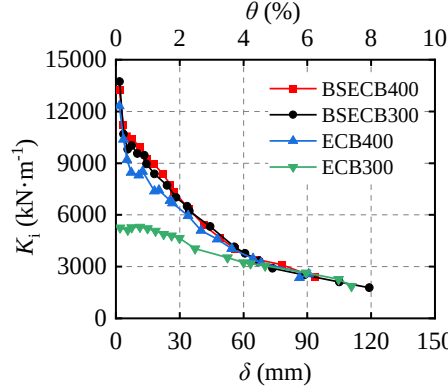


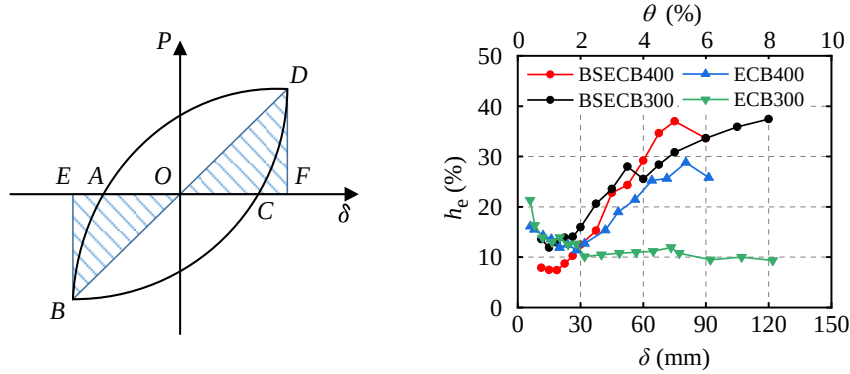
Fig.7 Stiffness degradations of specimens

2.2.4 Energy dissipation capacity

The equivalent viscous damping coefficient h_e is adopted to evaluate the energy dissipation capacity of the specimens. According to JGJ/T 101-2015 [10], h_e is given by:

$$h_e = \frac{1}{2\pi} \cdot \frac{S_{ABCD}}{S_{OBE} + S_{ODF}} \quad (2)$$

where S_{ABCD} represents the area enclosed by the hysteresis loop, and S_{OBE} and S_{ODF} denote the areas of triangles OBE and ODF, respectively. (see Fig.8a).



(a) Calculation method of h_e

(b) h_e - δ curve

Fig.8 Equivalent viscous damping coefficient curves

As shown in Fig.8b, except for the initial elastic stage (θ less than about 1%), the equivalent viscous damping coefficient (h_e) of specimens ECB400, BSECB300, and BSECB400 generally increases with the progression of loading, whereas h_e for ECB300 remains nearly constant throughout the loading process.

It can be observed that the energy dissipation capacities of BSECB300 and BSECB400 are

relatively high, reaching approximately 0.34 at $\theta = 6\%$, which is slightly greater than that of ECB400 (approximately 0.26 at $\theta = 6\%$). In contrast, specimen ECB300 exhibits significantly lower energy dissipation capacity, with h_e of approximately 0.10 at $\theta = 6\%$, compared with the other specimens.

3. FE modelling

3.1 Element types and interactions

A three-dimensional finite element (FE) model was developed using ABAQUS to simulate the behavior of beam-strengthened embedded CFST column base connections under cyclic loading. Owing to geometric symmetry, only half of each specimen was modeled. As shown in Fig.9, the concrete foundation and strengthening beams were modeled using 8-node solid elements with reduced integration (C3D8R), while the steel tube and reinforcing bars were represented by 4-node shell elements (S4R) and 2-node truss elements (T3D2), respectively. Based on a mesh convergence study, an element size of 50 mm, with local refinement to 25 mm, was adopted for the concrete foundation, while a uniform element size of 25 mm was used for the remaining components.

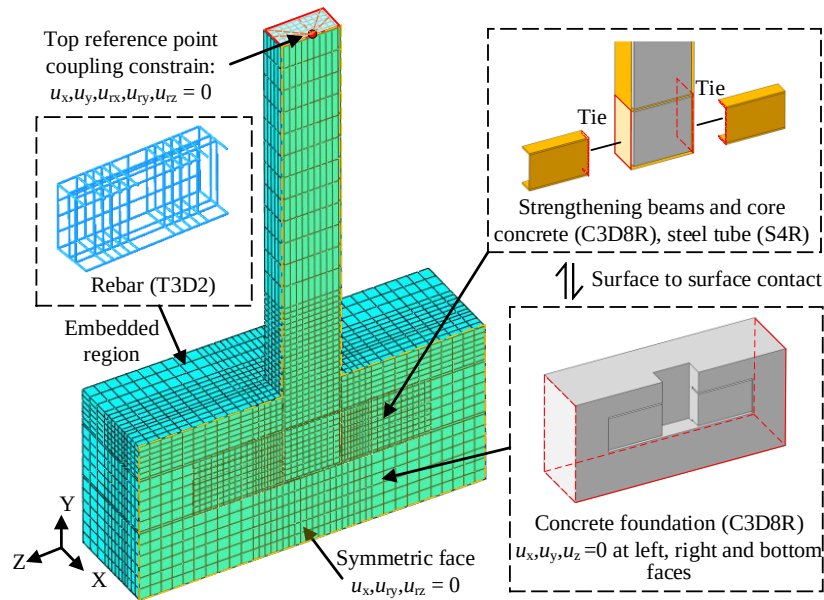


Fig.9 Details of FE model

The interaction relationships among the components of the FE model are illustrated in Fig.9. The connections between the strengthening beams and the steel tube were modeled using tie constraints. Surface-to-surface contact was employed to simulate the interactions between the strengthening beams and the surrounding concrete, as well as between the steel tube and the

concrete. For these contact interfaces, hard contact was defined in the normal direction, while a penalty friction formulation was adopted in the tangential direction. Previous studies [11] have indicated that the friction coefficient (μ) between steel and concrete typically ranges from 0.1 to 0.6; in this study, a value of 0.4 was adopted in accordance with GB 51022-2015 [12]. In addition, the bond behavior between reinforcing bars and concrete was simulated using the embedded region constraint.

The boundary conditions of the FE model are also shown in Fig.9. The left, right, and bottom surfaces of the concrete foundation were fully restrained. At the plane of symmetry, translational displacement in the X -direction and rotations about the Y - and Z -axes were constrained. Furthermore, displacement-controlled cyclic loading was applied at the top surface of the CFST column through a rigid kinematic coupling constraint.

3.2 Constitutive models of materials

3.2.1 Concrete modeling

The Concrete Damage Plasticity (CDP) model, with parameters listed in Table 2, was adopted to simulate the behavior of concrete. The compressive stress–strain relationships specified in GB 50010-2010 [13] and by Han *et al.* [14] were employed for plain concrete and confined concrete, respectively, to represent the foundation concrete and the concrete infilled within the steel tube. The tensile behavior of concrete was modeled using the constitutive relationship proposed by Shen *et al.* [15]. In FE modeling, the elastic modulus of concrete E_c is taken as $4700\sqrt{f'_c}$ [16], where f'_c is the characteristic compressive strength of concrete.

Table 2 Parameters of CDP model in ABAQUS

Dilation angle	Eccentricity	f_{b0}/f_{c0}	K	Viscosity Parameter
30	0.1	1.16	0.667	0.001

The stiffness degradation of concrete resulting from damage accumulation under cyclic loading was considered using the “focus point” model [17]. In this approach, two damage coefficients, d_c and d_t , are introduced to represent stiffness degradation under compressive and tensile loading, respectively [18].

3.2.2 Steel modeling

A combined hardening model, based on the von Mises yield criterion and the Prandtl–Reuss

flow rule, was adopted to simulate the cyclic behavior of steel. As summarized in Table 3, the key parameters of the model were determined in accordance with Shi *et al.* [19-20], where C_i and γ_i are material constants governing the evolution of the yield surface during plastic deformation.

Table 3 Parameters of the mixed hardening model of steel

Q_∞ (MPa)	b	C_1 (MPa)	γ_1	C_2 (MPa)	γ_2	C_3 (MPa)	γ_3
21	1.2	6013	173	5024	120	3026	32

A ductile damage model was employed to simulate the degradation of strength and stiffness induced by excessive plastic deformation. Within this framework, the damage behavior of steel is characterized by a damage initiation criterion and a subsequent damage evolution law (Fig.10). The damage variable (D_s) is defined as follows [21]:

$$D_s = 0.96(\bar{u}^{pl} / \bar{u}_f^{pl})^{2.42} \quad (3)$$

where $\bar{u}^{pl}$ and $\bar{u}_f^{pl}$ are the plastic displacement and plastic damage displacement in Eq.(3), [22], and $\bar{u}_f^{pl}$ is given by:

$$\bar{u}_f^{pl} = L_{char}(\bar{\varepsilon}_f^{pl} - \bar{\varepsilon}_D^{pl}) \quad (4)$$

where L_{char} is the element size, taken as 25mm in this study; $\bar{\varepsilon}_f^{pl}$ is taken as the failure plastic strain in the coupon tests; and $\bar{\varepsilon}_D^{pl}$ can be expressed as a function of the stress triaxiality [23].

The two extrem cases, $\bar{u}^{pl} = 0$ and $\bar{u}_f^{pl}$, correspond to no damage and complete damage of steel, respectively.

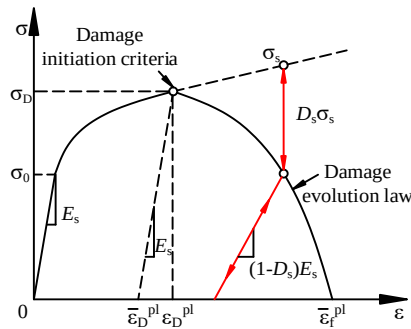


Fig.10 Ductile damage model of steel

3.3 Validation of FE modeling

The comparisons presented in Fig.5 demonstrate good agreements between the FE predictions and the experimental results. In particular, the pinching and sliding characteristics observed in the hysteretic responses of specimens ECB300 and ECB400 are well captured by the FE simulations. The peak loads of all specimens are summarized in Table 4, where the ratios

of the FE-predicted peak loads (P_{FEM}) to the experimental values (P_{Test}) range from 0.97 to 1.12, indicating satisfactory accuracy of the numerical model.

Table 4 Comparison between test and FE results

Specimens	Loading direction	P_{Test} (kN)	P_{FEM} (kN)	P_{FEM} / P_{Test}
ECB300	(+)	227.1	211.2	1.08
	(-)	234.0	209.2	1.12
ECB400	(+)	231.7	239.4	0.97
	(-)	254.4	239.4	1.06
BSECB300	(+)	240.9	249.0	0.97
	(-)	250.7	249.4	1.01
BSECB400	(+)	241.8	249.6	0.97
	(-)	244.9	248.1	0.99

The failure modes obtained from the FE analyses and experiments are presented in Fig.11 and Fig.12. As shown in Fig.11, the concrete damage observed in the tests is well reproduced in the FE simulations using the damage parameter DAMAGET. In addition, the local buckling of the CFST columns observed in specimens ECB400, BSECB300, and BSECB400 is accurately captured by the FE model (Fig.12).

These comparisons demonstrate that the proposed FE model is capable of reliably reproducing the key behavioral characteristics of the embedded column base connections investigated in this study, and it is therefore employed in the subsequent parametric analyses.

4. Parametric analyses

A parametric study was conducted using the validated FE model to evaluate the influence of key parameters on the behavior of BSECB connections. The parameters considered include the embedment depth (d_{embed}), axial load ratio (n), cross-sectional dimensions of the CFST column (H , B), length of the strengthening beams (L_s), width of the strengthening beam (b_s), web thickness of the strengthening beam (t_w), concrete strength (f_{cu}), and yield strength of the strengthening beam (f_{ys}).

The detailed parameters of all specimens considered in the parametric study are summarized in

As illustrated in Fig.18, the distributions of vertical stress in the concrete above and below the strengthening beams are similar for specimens with $L_s = 200$ mm, 250 mm, and 300 mm (Fig.18b-d), and these stress zones are noticeably wider than those for the specimen with $L_s = 150$ mm (Fig.18a). This observation suggests that increasing the beam length is effective for

improving the bending resistance of BSECB connections, while beyond a certain threshold this resistance improvement is very limited.

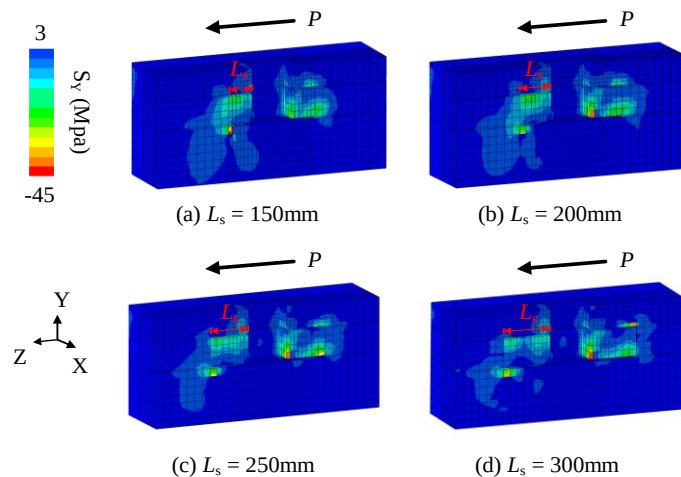


Fig.18 Vertical stress contours of the concrete foundations

4.1 Effect of strengthening beam width

As shown in Fig.19, increasing the width of the strengthening beam (b_s) enhances slightly the resistance of specimens with relatively shallow embedment depth ($d_{\text{embed}} = 250$ mm), as illustrated in Fig.19a. However, when the embedment depth is sufficiently large ($d_{\text{embed}} = 450$ mm), the influence of b_s on the resistance becomes negligible, as shown in Fig.19b.

4.2 Effect of web thickness of strengthening beam

The influence of the web thickness of the strengthening beam (t_w), which primarily governs

Table 5. Unless otherwise specified, the following baseline parameters were adopted: a concrete foundation with dimensions of 1460 mm (length) $\times$ 820 mm (width) $\times$ 700 mm (height); a CFST column with a steel tube section of 250 mm $\times$ 250 mm $\times$ 20 mm; strengthening beams with a cross-section of $h_s \times b_s \times t_w \times t_f = 200$ mm $\times$ 200 mm $\times$ 8 mm $\times$ 12 mm; a beam length (L_s) of 250 mm; and a yield strength of reinforcing steel of 235 MPa.

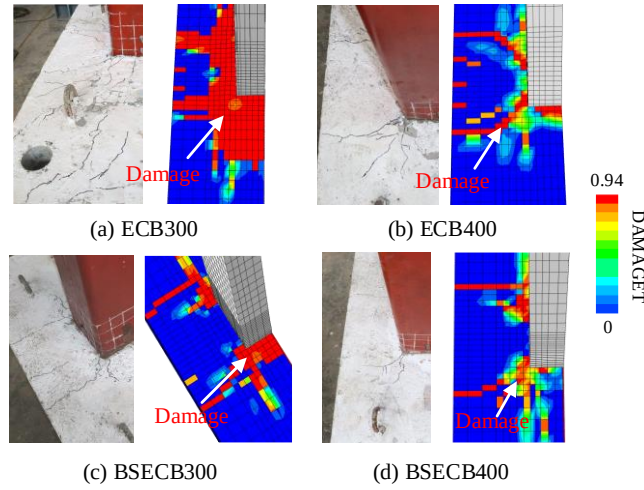


Fig.11 Comparison of concrete damage

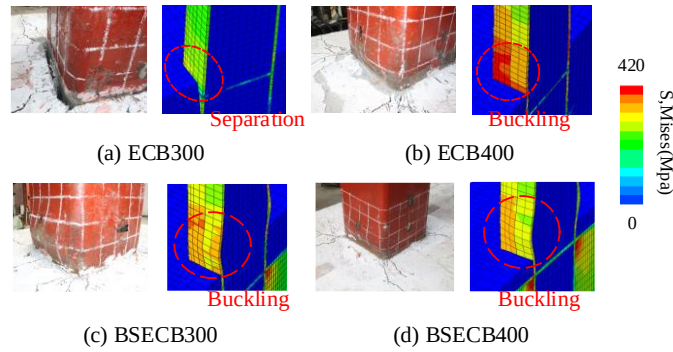


Fig.12 Comparison of CFST column failure

In addition, the diameter and arrangement of reinforcing bars in the concrete foundation were consistent with those used in the tested specimens (Fig.2), and lateral loads were applied at a height of 1500 mm above the foundation surface. It should be noted that a relatively compact CFST section was adopted in the parametric study to delay premature section failure of the column.

The axial load ratio (n), representing the axial compressive load applied to the CFST column, is defined in Eqs.(5)-(6):

$$n = \frac{N}{N_{sc}} \quad (5)$$

$$N_u = f_y A_s + 0.85 f'_c A_c \quad (6)$$

where N_u is the compressive resistance of CFST section [16]; and A_s and A_c are the areas of steel tube and concrete of CFST section.

4.3 Effect of embedment depth

The skeleton curves of Group G1 specimens with varying embedment depths (d_{embed}) are presented in Fig.13. It can be observed that both the ultimate load and initial stiffness of the

BSECB connections increase with increasing d_{embed} when the embedment depth does not exceed 450 mm. However, a comparison between the curves corresponding to $d_{\text{embed}} = 450$ mm and 550 mm indicates that further increasing the embedment depth results in no significant enhancement in either ultimate load or initial stiffness. This behavior can be attributed to a transition in the governing failure mode, from failure of the strengthening beams (for $d_{\text{embed}} = 250$ mm and 350 mm) to section failure of the CFST column at approximately $d_{\text{embed}} = 450$ mm, as illustrated in Fig.14.

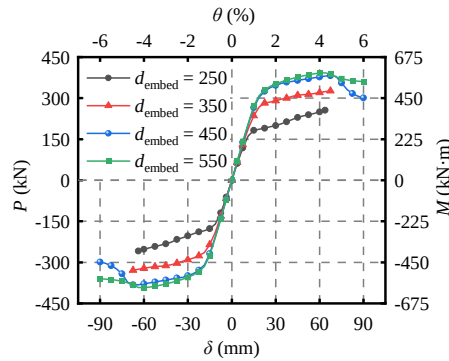


Fig.13 Effect of embedment depth (G1 specimens)

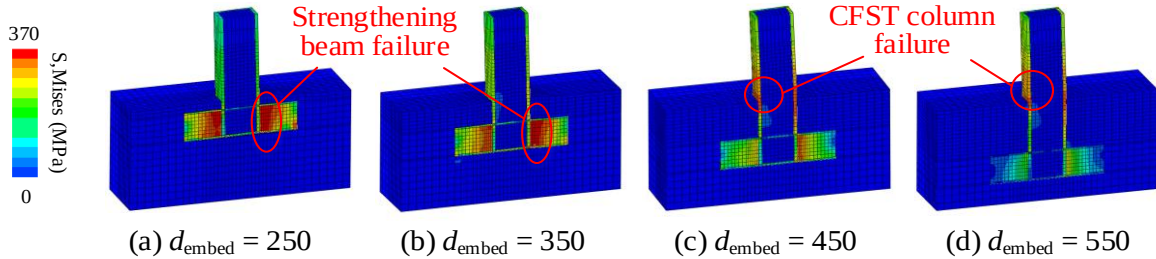


Fig.14 Mises stress contour of G1 specimens ($P=P_{\text{max}}$)

As shown in Fig.15, at the ultimate state, severe compressive damage is observed in the concrete above the strengthening beams for specimens with relatively small embedment depths (i.e., 250 mm and 350 mm). In contrast, for specimens with larger embedment depths (i.e., 450 mm and 550 mm), the compressive damage is primarily concentrated near the top surface of the foundation and remains localized in the regions surrounding the CFST column.

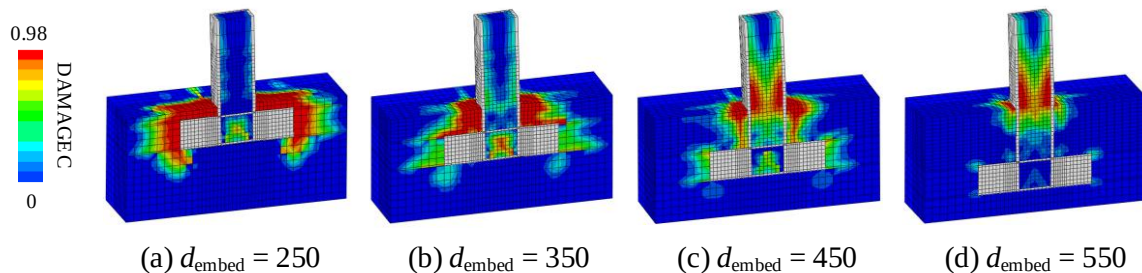


Fig.15 Concrete damage of G1 specimens ($P = P_{\text{max}}$)

4.4 Effect of axial load

The influence of axial compressive load on the behavior of BSECB connections (Group G2 specimens) is illustrated in Fig.16, where two embedment depths, 250 mm and 450 mm, respectively representing relatively shallow and deep embedment conditions, are considered. As shown in Fig.16a, the presence of axial compressive force appears to enhance the lateral resistance of BSECB connections with shallow embedment ($d_{\text{embed}} = 250$ mm), as the ultimate loads corresponding to $n = 0.2$ and 0.4 exceed that of $n = 0$. In contrast, for specimens with $d_{\text{embed}} = 450$ mm (Fig.16b), the lateral resistance decreases with increasing axial load ratio n . This difference can be attributed to the governing failure mechanism. For shallow embedment, axial compression improves confinement and load transfer, thereby enhancing resistance. However, for deeper embedment, the failure is governed by the strength of the CFST section, and the presence of axial force reduces its bending capacity, consistent with the provisions of Eurocode 4 [16].

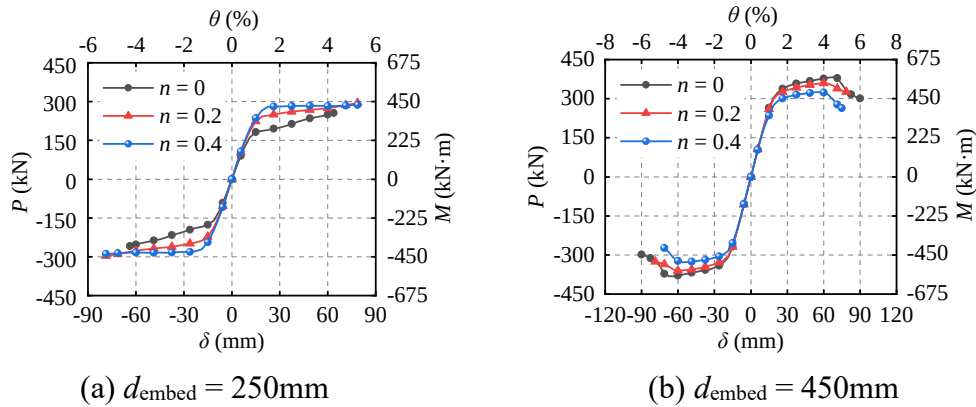


Fig.16 Effects of axial load on skeleton curves (G2 specimens)

4.5 Effect of strengthening beam length

The influence of strengthening beam length (L_s) is illustrated in Fig.17 for specimens with relatively shallow embedment depths ($d_{\text{embed}} = 250$ mm and 350 mm), corresponding to Group G3 in As illustrated in Fig.18, the distributions of vertical stress in the concrete above and below the strengthening beams are similar for specimens with $L_s = 200$ mm, 250 mm, and 300 mm (Fig.18b-d), and these stress zones are noticeably wider than those for the specimen with $L_s = 150$ mm (Fig.18a). This observation suggests that increasing the beam length is effective for

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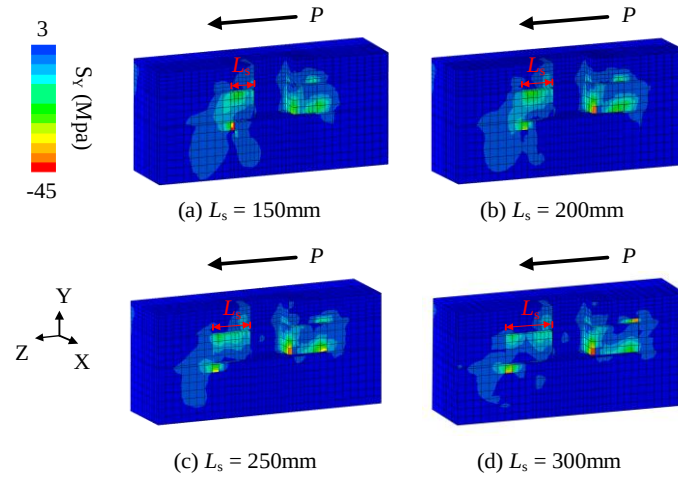


Fig.18 Vertical stress contours of the concrete foundations

4.6 Effect of strengthening beam width

As shown in Fig.19, increasing the width of the strengthening beam (b_s) enhances slightly the resistance of specimens with relatively shallow embedment depth ($d_{\text{embed}} = 250$ mm), as illustrated in Fig.19a. However, when the embedment depth is sufficiently large ($d_{\text{embed}} = 450$ mm), the influence of b_s on the resistance becomes negligible, as shown in Fig.19b.

4.7 Effect of web thickness of strengthening beam

The influence of the web thickness of the strengthening beam (t_w), which primarily governs

Table 5. As shown in Fig.17a, the lateral resistance of specimens with $d_{\text{embed}} = 250$ mm increases with increasing L_s . However, this enhancement becomes less pronounced for specimens with a larger embedment depth ($d_{\text{embed}} = 350$ mm), as shown in Fig.17b. In contrast, the comparisons in Fig.17 indicate that the effect of L_s on the initial stiffness is negligible.

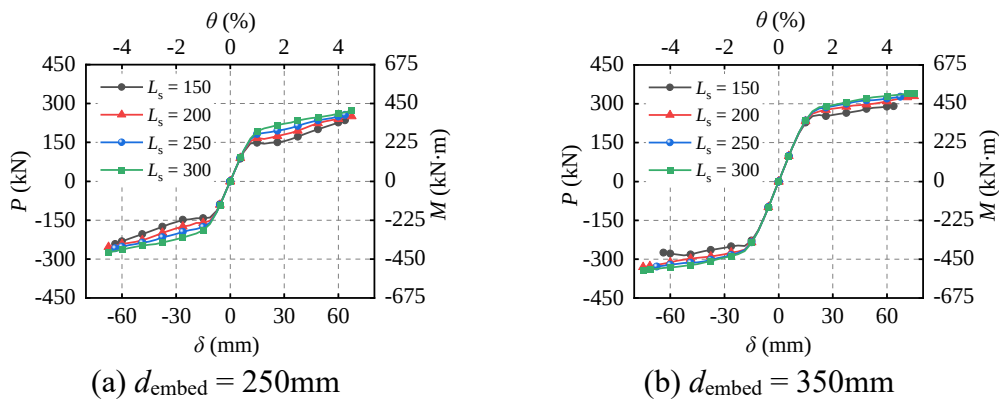


Fig.17 Effect of strengthening beam length on skeleton curves (G3 specimens)

As illustrated in Fig.18, the distributions of vertical stress in the concrete above and below the strengthening beams are similar for specimens with $L_s = 200$ mm, 250 mm, and 300 mm (Fig.18b-d), and these stress zones are noticeably wider than those for the specimen with $L_s = 150$ mm (Fig.18a). This observation suggests that increasing the beam length is effective for improving the bending resistance of BSECB connections, while beyond a certain threshold this resistance improvement is very limited.

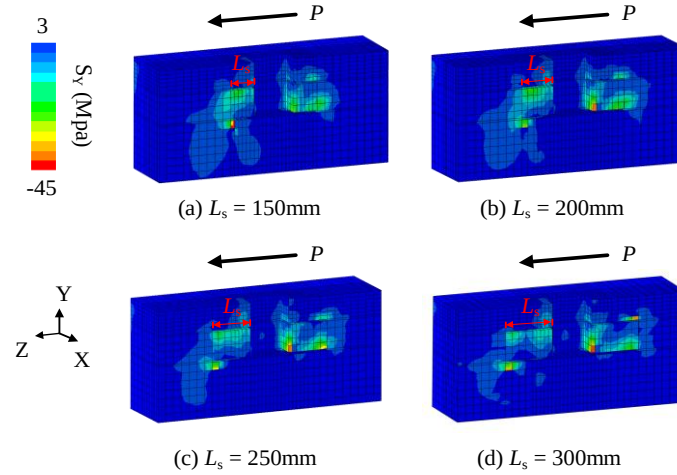


Fig.18 Vertical stress contours of the concrete foundations

4.8 Effect of strengthening beam width

As shown in Fig.19, increasing the width of the strengthening beam (b_s) enhances slightly the resistance of specimens with relatively shallow embedment depth ($d_{\text{embed}} = 250$ mm), as illustrated in Fig.19a. However, when the embedment depth is sufficiently large ($d_{\text{embed}} = 450$ mm), the influence of b_s on the resistance becomes negligible, as shown in Fig.19b.

4.9 Effect of web thickness of strengthening beam

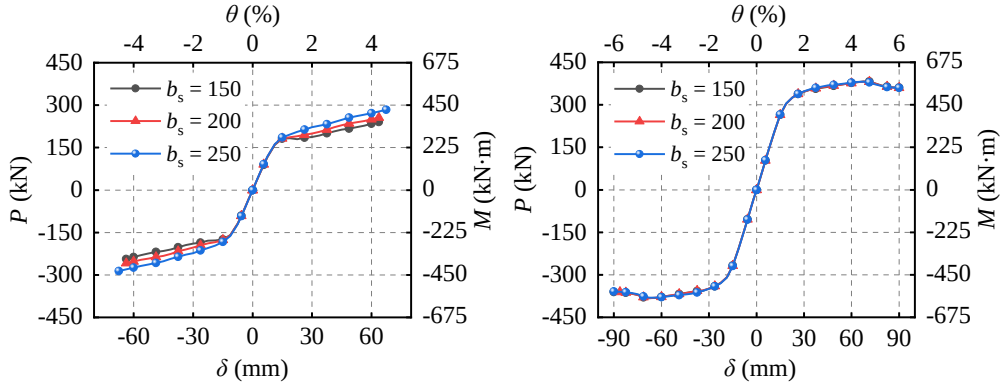
The influence of the web thickness of the strengthening beam (t_w), which primarily governs

Table 5 Details of specimens in parametric study

Series	Specimens	d_{embed} (mm)	n	t (mm)	$H=B$ (mm)	L_s (mm)	b_s (mm)	t_w (mm)	f_{cu} (MPa)	f_{ys} (MPa)
G1	d-250	250	0	20	250	250	200	8	35	235
	d-350	350								
	d-450	450								
	d-550	550								
G2	d-250-n-0	250	0	20	250	250	200	8	35	235
	d-250-n-0.2		0.2							
	d-250-n-0.4		0.4							
	d-350-n-0	350	0	20	250	250	200	8	35	235
	d-350-n-0.2		0.2							
	d-350-n-0.4		0.4							
	d-450-n-0	450	0	20	250	250	200	8	35	235
	d-450-n-0.2		0.2							
	d-450-n-0.4		0.4							
G3	d-250- L_s -150	250	0	20	250	150	200	8	35	235
	d-250- L_s -200					200				

	<i>d</i> -250- <i>L_s</i> -250					250				
	<i>d</i> -250- <i>L_s</i> -300					300				
	<i>d</i> -350- <i>L_s</i> -150	350	0	20	250	150	200	8	35	235
	<i>d</i> -350- <i>L_s</i> -200					200				
	<i>d</i> -350- <i>L_s</i> -250					250				
	<i>d</i> -350- <i>L_s</i> -300					300				
	<i>d</i> -450- <i>L_s</i> -150	450	0	20	250	150	200	8	35	235
	<i>d</i> -450- <i>L_s</i> -200					200				
	<i>d</i> -450- <i>L_s</i> -250					250				
	<i>d</i> -450- <i>L_s</i> -300					300				
G4	<i>d</i> -250- <i>b_s</i> -150	250	0	20	250	250	150	8	35	235
	<i>d</i> -250- <i>b_s</i> -200						200			
	<i>d</i> -250- <i>b_s</i> -250						250			
	<i>d</i> -350- <i>b_s</i> -150	350	0	20	250	250	150	8	35	235
	<i>d</i> -350- <i>b_s</i> -200						200			
	<i>d</i> -350- <i>b_s</i> -250						250			
	<i>d</i> -450- <i>b_s</i> -150	450	0	20	250	250	150	8	35	235
	<i>d</i> -450- <i>b_s</i> -200						200			
G5	<i>d</i> -450- <i>b_s</i> -250						250			
	<i>d</i> -250- <i>t_w</i> -4	250	0	20	250	250	200	4	35	235
	<i>d</i> -250- <i>t_w</i> -8							8		
	<i>d</i> -250- <i>t_w</i> -16							16		
	<i>d</i> -350- <i>t_w</i> -4	350	0	20	250	250	200	4	35	235
	<i>d</i> -350- <i>t_w</i> -8							8		
	<i>d</i> -350- <i>t_w</i> -16							16		
	<i>d</i> -450- <i>t_w</i> -4	450	0	20	250	250	200	4	35	235
G6	<i>d</i> -450- <i>t_w</i> -8							8		
	<i>d</i> -450- <i>t_w</i> -16							16		
	<i>d</i> -250- <i>f_{cu}</i> -35	250	0	20	250	250	200	8	35	235
	<i>d</i> -250- <i>f_{cu}</i> -50								50	
	<i>d</i> -250- <i>f_{cu}</i> -60								60	
	<i>d</i> -350- <i>f_{cu}</i> -35	350	0	20	250	250	200	8	35	235
	<i>d</i> -350- <i>f_{cu}</i> -50								50	
	<i>d</i> -350- <i>f_{cu}</i> -60								60	
G7	<i>d</i> -450- <i>f_{cu}</i> -35	450	0	20	250	250	200	8	35	235
	<i>d</i> -450- <i>f_{cu}</i> -50								50	
	<i>d</i> -450- <i>f_{cu}</i> -60								60	
	<i>d</i> -250- <i>f_{ys}</i> -235	250	0	20	250	250	200	8	35	235
	<i>d</i> -250- <i>f_{ys}</i> -355									355
	<i>d</i> -250- <i>f_{ys}</i> -390									390
	<i>d</i> -250- <i>f_{ys}</i> -420									420
	<i>d</i> -350- <i>f_{ys}</i> -235	350	0	20	250	250	200	8	35	235
	<i>d</i> -350- <i>f_{ys}</i> -355									355
	<i>d</i> -350- <i>f_{ys}</i> -390									390
	<i>d</i> -350- <i>f_{ys}</i> -420									420
	<i>d</i> -450- <i>f_{ys}</i> -235	450	0	20	250	250	200	8	35	235
	<i>d</i> -450- <i>f_{ys}</i> -355									355
	<i>d</i> -450- <i>f_{ys}</i> -390									390
	<i>d</i> -450- <i>f_{ys}</i> -420									420

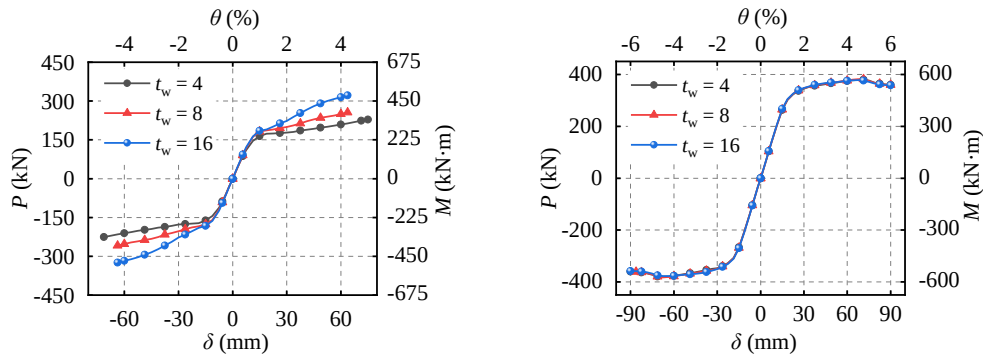
its shear capacity, on the behavior of BSECB connections is illustrated in Fig.20. As shown in Fig.20a, increasing t_w enhances the lateral resistance of connections with shallow embedment ($d_{embed} = 250$ mm). However, for connections with relatively deep embedment ($d_{embed} = 450$ mm), the effect of increasing t_w becomes negligible, as shown in Fig.20b. This is because, under such conditions, the resistance of the BSECB connection is governed by the strength of the CFST section rather than the strengthening beam.



(a) $d_{\text{embed}} = 250\text{mm}$

(b) $d_{\text{embed}} = 450\text{mm}$

Fig.19 Effect of flange width on skeleton curves (G4 specimens)



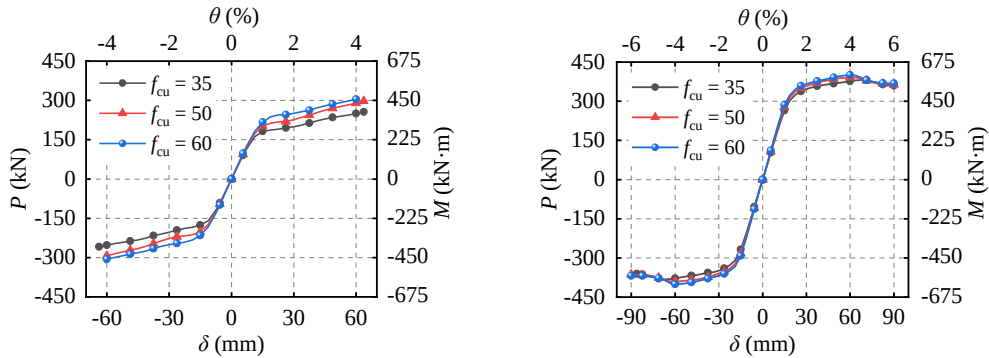
(a) $d_{\text{embed}} = 250\text{mm}$

(b) $d_{\text{embed}} = 450\text{mm}$

Fig.20 Effect of web thickness of strengthening beam (G5 specimens)

4.10 Effect of concrete strength

As shown in Fig.21a, for BSECB connections with shallow embedment ($d_{\text{embed}} = 250\text{ mm}$), increasing the concrete strength enhances the lateral resistance of the connection. This improvement can be attributed to the higher bearing capacity of concrete, which provides greater moment resistance in the regions surrounding the strengthening beams and the CFST section. In contrast, as illustrated in Fig.21b, the influence of concrete strength on the behavior of BSECB connections with relatively deep embedment ($d_{\text{embed}} = 450\text{ mm}$) is limited.



(a) $d_{\text{embed}} = 250\text{mm}$

(b) $d_{\text{embed}} = 450\text{mm}$

Fig.21 Effect of concrete strength on skeleton curves (G6 specimens)

4.11 Effect of strengthening beam strength

Fig.22 presents the skeleton curves of BSECB connections with varying yield strengths of the strengthening beams (f_{ys}). For connections with shallow embedment, the use of higher-strength steel in the strengthening beams leads to an increase in lateral resistance, as shown in **Fig.22a**. In contrast, for connections with relatively deep embedment, this effect becomes marginal, as illustrated in **Fig.22b**.

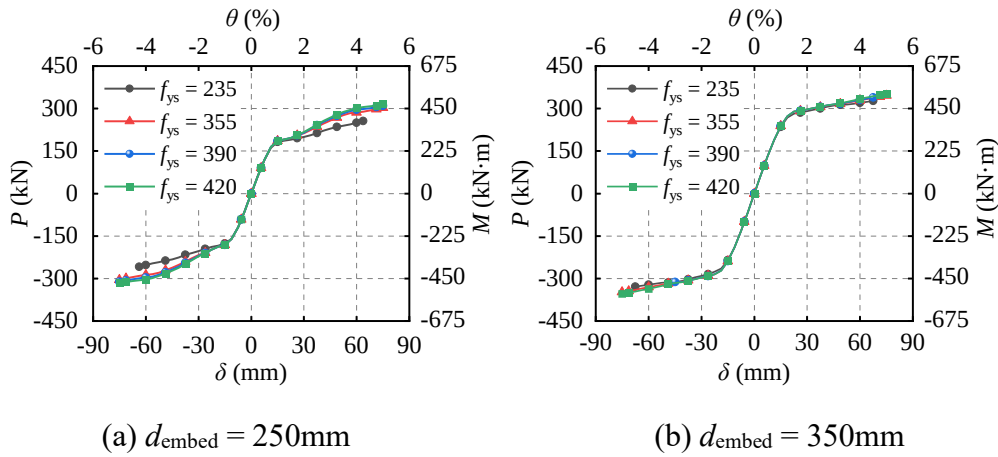


Fig.22 Effect of strengthening beam strength on skeleton curves (G7 specimens)

5. Ultimate load prediction

5.1 Potential ultimate conditions

The strength of embedded CFST column base connections may be governed by three potential failure modes: CFST column failure, concrete bearing failure, and upward punching failure of the concrete foundation. The corresponding ultimate moments are denoted as M_y , M_{fb} and M_{pun} , respectively. Accordingly, the ultimate moment capacity of BSECB connections (M_{pre}) can be expressed as:

$$M_{pre} = \min(M_y, M_{fb}, M_{pun}) \quad (7)$$

Since the flexural strength of the CFST column (M_y) can be determined in accordance with Eurocode 4 [16], this study focuses on the development of analytical models for predicting M_{fb} and M_{pun} . For clarity, all external loads and resultant sectional forces are evaluated with respect to a reference point defined as the centroid of the column section at the top surface of the concrete foundation (see **Fig.23**).

5.2 Concrete bearing failure

When the embedment depth is insufficient, the strength of the embedded column base may be governed by bearing failure of the concrete surrounding the CFST column (Fig.23). For the BSECB connections considered in this study, the contribution of the strengthening beams should be incorporated in addition to the bearing resistance provided by the surrounding concrete.

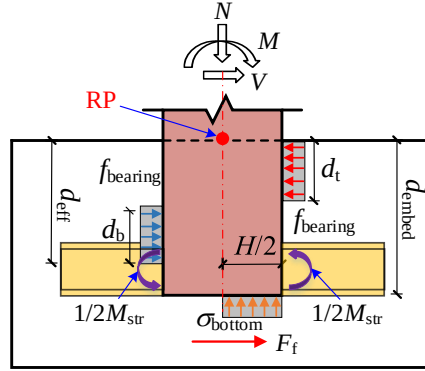


Fig.23 Ultimate state of bearing failure mode

As reported in [3], the bearing stress in the surrounding concrete of embedded steel column base connections decreases with increasing embedment depth, indicating that only a limited portion of the embedment is effective in resisting shear force and bending moment. Accordingly, an effective embedment depth (d_{eff}) is defined for the bearing stress analysis, as given by Eqs.(8)-(9) ([3] and [24]):

$$d_{eff} = \frac{C_0}{\rho} \leq d_{embed} \quad (8)$$

$$\rho = \left[\frac{E_c}{4(EI)_{section}} \right]^{1/4} \quad (9)$$

in which E_c are the Young's modulus of the concrete; $(EI)_{section}$ is the bending rigidity of CFST column; and C_0 is a constant and can be taken as $\pi/2$ according to Xu *et.al* [25].

The friction force, F_f , at the bottom end of the column is caused by the axial compressive force (N), where the friction coefficient μ is taken as 0.4 according to GB 51022-2015 [12].

$$F_f = \mu N \quad (10)$$

As illustrated in Fig.23, the force and bending moment balances lead to:

$$V + F_f = f_{bearing} B(d_t - d_b) \quad (11)$$

$$N = f_{\text{bottom}} BH / 2 \quad (12)$$

$$M - M_{\text{str}} + f_{\text{bearing}} B \frac{d_t^2}{2} = F_f d_{\text{embed}} + f_{\text{bearing}} B d_b \left(d_t + \frac{d_b}{2} \right) + N \frac{H}{4} \quad (13)$$

where M_{str} is the bending moment from the contribution of strengthening beams.

It should be noted that, in Eq.(13), the compressive width at the base of the CFST column under axial load at the ultimate state is assumed to be $H/2$. The ultimate shear force (V_u) and bending moment (M_u) of the column base connection without strengthening beams can be determined using Eq.(14) and Eq.(15), respectively, where the relationships $d_t = d_{\text{eff}}$ and $d_t = d_b = 0.5d_{\text{eff}}$ are adopted. It is further noted that f_{bottom} and F_f reduce to zero when the axial load (N) is not considered.

$$V_u = f_{\text{bearing}} B d_{\text{eff}} \quad (14)$$

$$M_u = 0.25 V_u d_{\text{eff}} \quad (15)$$

where f_{bearing} is the bearing strength of concrete, which, according to GB 50010-2010 [13] and Eurocode 2 [26], can be expressed as.

$$f_{\text{bearing}} = 1.35 \beta_c \beta_l f_{\text{ck}} \quad (16)$$

$$\beta_l = \sqrt{\frac{A_b}{A_l}} \quad (17)$$

where β_c is a reduction factor of concrete strength, and can be taken as 1.0 for $f_{\text{cu}} \leq 50$ MPa; β_l is the strength enhancement factor for the localized compression; A_b is the loading area; A_l is the design area in the similar shape of A_b . According to [13], for the considered case, β_l can be taken as 1.7; and f_{ck} is the characteristic compressive stress of concrete and can be taken as 0.67 f_{cu} .

Based on Eqs.(10)-(17), the V - M relationship with the axial load N taken into account can be given by:

$$\left(\frac{V + \mu N}{V_u} \right)^2 + \frac{M - M_{\text{str}} + 0.5 V d_{\text{eff}} - N (0.25 H + \mu d_{\text{embed}} - 0.5 \mu d_{\text{eff}})}{M_u} = 1 \quad (18)$$

in which M_{str} may be controlled by four possible failure modes:

$$M_{\text{str}} = \min(M_{\text{CB}}, M_{\text{BS}}, M_{\text{BB}}, M_{\text{CS}}) \quad (19)$$

in which M_{CB} , M_{BS} , M_{BB} and M_{CS} represent the bending moments of strengthening beams failed by the bearing failure of surrounding concrete, strengthening beam shear failure, strengthening

beam bending failure and panel zone shear failure, respectively. The ultimate moment corresponding to the concrete bearing failure mode M_{fb} is determined from Eq.(18) by considering the geometry, material strengths, axial load, friction, and M_{str} .

5.2.1 Moment at surrounding concrete bearing failure

As illustrated in Fig.24, the bearing stresses developed in the concrete above and below the strengthening beams contribute additional bending resistance to the BSECB connections. As discussed in the previous section, the embedded portion of the CFST column may not be fully effective in resisting bending moment; therefore, an effective embedment length is defined in Eqs.(20)-(21). Similarly, the effective length of the strengthening beam is defined as:

$$L_{s,eff} = \frac{C_0}{\rho_s} \leq L_s \quad (20)$$

$$\rho_s = \left(\frac{E_c}{4E_s I_b} \right)^{1/4} \quad (21)$$

where E_s is the elastic modulus of steel and I_b is the moment of inertia of strengthening beam.

Based on the stress distribution illustrated in Fig.18, the resultant bending moment contributed by the bearing stresses in the concrete surrounding the strengthening beams (M_{CB}) can be expressed as:

$$M_{CB} = f_{ck} b_s L_{s,eff} (H + L_{s,eff}) \quad (22)$$

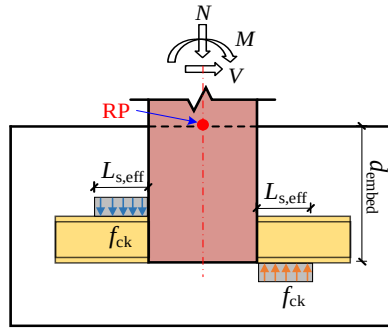


Fig.24 Mechanism of surrounding concrete failure

5.2.2 Moment at strengthening beam shear failure

The strength of the strengthening beams may be governed by shear failure of the cross-section (Fig.25). For strengthening beams with I-sections, the shear capacity of the cross-section can be expressed as:

$$V_{BS} = A_{sw} f_v = 0.577 t_w h_s f_{ys} \quad (23)$$

in which A_{sw} , h_s and t_w are the area, height and web thickness of strengthening beams,

respectively; and f_{ys} is the yield strength of steel of strengthening beam.

The contribution of the concrete infill between the flanges of the strengthening beam can be evaluated using the strut-and-tie model illustrated in Fig.25b, which is adapted from Pertold [27]. According to Pertold [27], the lever arm (z) associated with the shear forces can be expressed as:

$$z = 0.6L_s \quad (24)$$

Then, the angle of inclination of the concrete strut component is

$$\theta = \arctan\left(\frac{0.6L_s}{h_{sw}}\right) \quad (25)$$

where h_{sw} is the web height of strengthening beam section.

The effective area of concrete strut section, A_{cc} , can be obtained with

$$A_{cc} = 0.8h_{sw} \sin \theta (b_s - t_w) \quad (26)$$

The effective compressive strength of concrete strut, f_{ce} , shall be calculated using ACI 318-19 [28]:

$$f_{ce} = 0.85\beta_c\beta_s f'_c \quad (27)$$

where the two coefficients β_c and β_s are taken as 1.0 and 0.75 according to ACI 318-19 [28].

The vertical shear force, V_{BCC} , from the concrete strut is given by:

$$V_{BCC} = A_{cc} f_{ce} \cos \theta \quad (28)$$

Assuming the compressive stresses acting on the strengthening beams are uniform in the effective length, the bending moment is given as

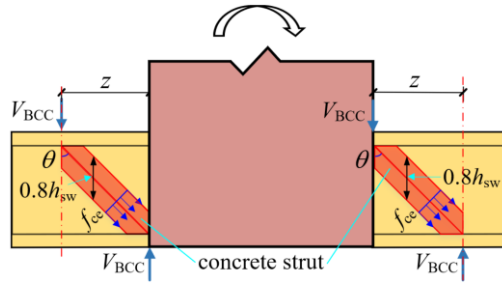
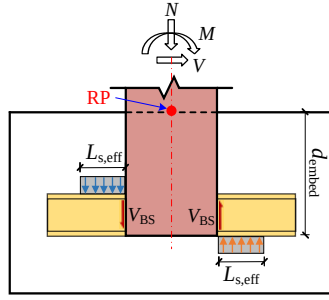
$$M_{BS} = V_{BS} (H + L_{s,eff}) + 1.2V_{BCC}L_s \quad (29)$$

5.2.3 Moment at strengthening beam bending failure

As illustrated in Fig.26, the contribution of the strengthening beams may be governed by the flexural capacity of the cross-section at their fixed ends. In this case, the maximum bending moment provided by the strengthening beams is given by:

$$M_{BB} = 2W_p f_{ys} \quad (30)$$

where W_p is the plastic section modulus of the strengthening beam.



(a) Shear force of beam web

(b) Shear force due to concrete strut

Fig.25 Mechanism of web shear failure

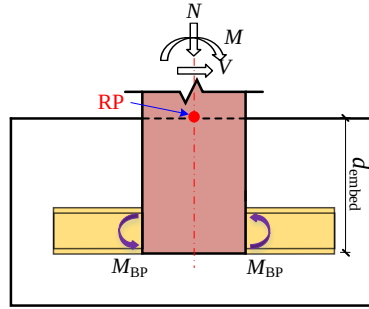


Fig.26 Bending failure of strengthening beams

5.2.4 Moment at joint zone shear failure

Under the bending action induced by the strengthening beams, the CFST column may fail due to shear deformation of the joint zone (Fig.27). The shear resistance of the CFST section is assumed to be provided by the side walls of the steel tube (Fig.27a) and the infilled concrete (Fig.27b). The shear resistance of steel tube can be expressed as:

$$V_{CS} = 2t(H - 2t)f_v \quad (31)$$

where H is the height of CFST section; t is thickness of side walls of steel tube; and f_v is the shear strength of steel.

As illustrated in [29], the effective area of concrete strut section in steel beam to CFST column joint with diaphragms, A_{co} , can be given by

$$A_{co} = 0.3(B - 2t)\sqrt{(H - 2t)^2 + (h_s - 2t_f)^2} \quad (32)$$

Then, the horizontal component of the compressive force of the concrete strut of the joint zone, N_{ch} , is given by

$$N_{ch} = f_{ck}A_{co} \cos \alpha \quad (33)$$

where α is the angle between the concrete strut and horizontal axis (Fig.27b). Substituting

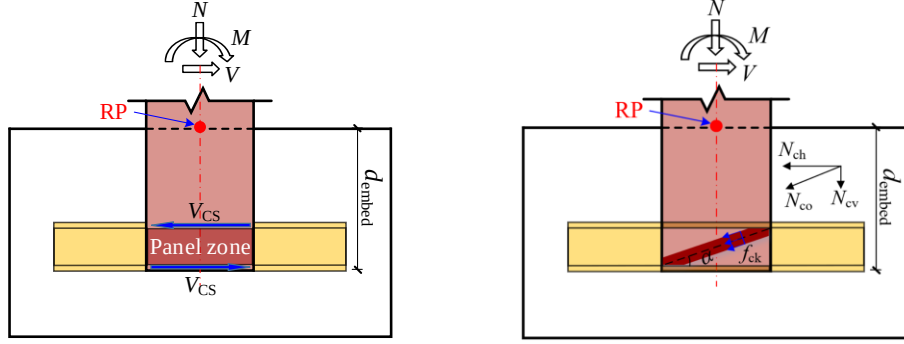
Eq.(32) into Eq.(33) and using the relationship $\cos \alpha = (H - 2t) / \sqrt{(H - 2t)^2 + (h_s - 2t_f)^2}$

lead to:

$$N_{ch} = 0.3 f_{ck} (B - 2t)(H - 2t) \quad (34)$$

Then, the resultant bending moment (M_{CS}) induced by V_{CS} and N_{ch} can be obtained as

$$M_{CS} = (V_{CS} + N_{ch})(h_s - t_f) \quad (35)$$



(a) Shear force of side walls

(b) Shear force of concrete strut

Fig.27 Mechanism of column shear failure

5.3 Upward punching failure of concrete foundation

With the inclusion of strengthening beams, the upward punching resistance of BSECB connections is enhanced due to the enlargement of the effective punching area (Fig.28). As illustrated in Fig.28, the punching area can be divided into two regions, where A_{pb} denotes the additional area contributed by the strengthening beams.

Accordingly, the corresponding vertical punching forces, F_{pun}^a and F_{pun}^b , are associated with the punching areas A_{pa} and A_{pb} , respectively. From the schematic diagram shown in Fig.28, the two areas can be given by:

$$A_{pa} = d_{embed} b_s + \frac{1}{2} \pi d_{embed}^2 \quad (36)$$

$$A_{pb} = 2d_{embed} L_s \quad (37)$$

It should be noted that A_{pa} and A_{pb} in Eqs.(36)-(37) are the projected areas, and in calculating these two areas, their outer edges should not beyond the concrete foundation. Then, the punching forces corresponding to these two areas are:

$$F_{pun}^a = 0.7 \beta_h f_t A_{pa} + 0.8 f_{yv} n_{pa} A_s \quad (38)$$

$$F_{pun}^b = 0.7 \beta_h f_t A_{pb} + 0.8 f_{yv} n_{pb} A_s \quad (39)$$

where n_{pa} and n_{pb} are the stirrup numbers within the projected areas A_{pa} and A_{pb} , respectively; and A_s is the area of a stirrup section.

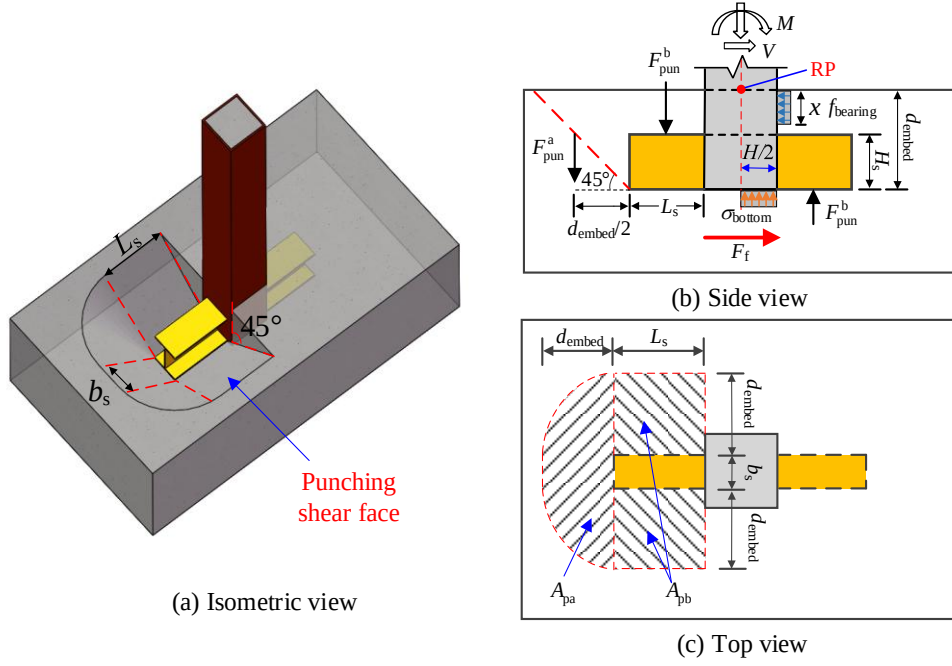


Fig.28 Mechanism of the second punching shear mode

As illustrated in Fig.28, using the expression of the friction force at the bottom end of CFST column $F_f = \mu(N + F_{pun}^b + F_{pun}^a)$, the force balance in the horizontal direction leads to:

$$x = \frac{V + \mu(N + F_{pun}^a + F_{pun}^b)}{f_{bearing} B} \quad (40)$$

where x is the width of the concrete bearing area, and B is the width of column section.

The equilibrium condition of the bending moment with respect to the reference point (RP in Fig.28) gives:

$$M_{pun} = \frac{F_{pun}^a d_{embed} + F_{pun} (2L_s + H) - f_{bearing} Bx^2}{2} + \frac{(N + F_{pun}) (4\mu d_{embed} + H) + F_{pun}^b H}{4} \quad (41)$$

where $F_{pun} = F_{pun}^a + F_{pun}^b$.

5.4 Comparison of results

Comparisons between the proposed predictions (M_{pre}) and the FE results (M_{FEM}) are presented in Table 6 using numerical 64 examples given in Table 5. The minimum bending contribution of the strengthening beams (M_{str}), governed by various potential failure modes (i.e. M_{CB} , M_{BB} , M_{BS} , M_{CS}), is also summarized. In Table 6, M_{fb} denotes the bending moment calculated using Eq.(18) incorporating M_{str} , while M_{pun} is obtained from Eq.(41). Accordingly, the bending moment capacity of a BSECB connection (M_{pre}) is determined as the minimum value among M_{fb} , M_{pun} , and M_y (the bending capacity of the CFST column).

G1	<i>d</i> -250	374.5	585.2	179.1	241.2	300.0	179.1	358.0	557.3	673.6	BS	358.0	0.96
	<i>d</i> -350	480.3	585.2	179.1	241.2	300.0	179.1	526.6	557.3	1422.3	BS	526.6	1.10
	<i>d</i> -450	573.5	585.2	179.1	241.2	300.0	179.1	701.3	557.3	1692.2	SS	557.3	0.97
	<i>d</i> -550	589.0	585.2	179.1	241.2	300.0	179.1	701.3	557.3	1842.9	SS	557.3	0.95
G2	<i>d</i> -250- <i>n</i> -0	374.5	585.2	179.1	241.2	300.0	179.1	358.0	557.3	673.6	BS	358.0	0.96
	<i>d</i> -250- <i>n</i> -0.2	411.7	585.2	179.1	241.2	300.0	179.1	461.0	516.8	803.6	BS	461.0	1.12
	<i>d</i> -250- <i>n</i> -0.4	426.2	585.2	179.1	241.2	300.0	179.1	555.7	470.3	915.4	SS	470.3	1.10
	<i>d</i> -350- <i>n</i> -0	480.3	585.2	179.1	241.2	300.0	179.1	526.6	557.3	1422.3	BS	526.6	1.10
	<i>d</i> -350- <i>n</i> -0.2	508.3	585.2	179.1	241.2	300.0	179.1	643.6	515.8	1558.5	SS	515.8	1.01
	<i>d</i> -350- <i>n</i> -0.4	478.7	585.2	179.1	241.2	300.0	179.1	752.3	469.2	1677.1	SS	469.2	0.98
	<i>d</i> -450- <i>n</i> -0	573.5	585.2	179.1	241.2	300.0	179.1	701.3	557.3	1692.2	SS	557.3	0.97
	<i>d</i> -450- <i>n</i> -0.2	538.4	585.2	179.1	241.2	300.0	179.1	835.2	515.8	1857.6	SS	515.8	0.96
G3	<i>d</i> -450- <i>n</i> -0.4	488.0	585.2	179.1	241.2	300.0	179.1	960.8	469.2	2004.3	SS	469.2	0.96
	<i>d</i> -250- <i>L_s</i> -150	339.2	280.9	121.6	241.2	300.0	121.6	305.2	557.3	529.2	BS	305.2	0.90
	<i>d</i> -250- <i>L_s</i> -200	361.1	421.3	150.9	241.2	300.0	150.9	332.1	557.3	595.7	BS	332.1	0.92
	<i>d</i> -250- <i>L_s</i> -250	374.5	585.2	179.1	241.2	300.0	179.1	358.0	557.3	673.6	BS	358.0	0.96
	<i>d</i> -250- <i>L_s</i> -300	391.7	772.5	205.1	241.2	300.0	205.1	381.9	557.3	825.5	BS	381.9	0.97
	<i>d</i> -350- <i>L_s</i> -150	432.8	280.9	121.6	241.2	300.0	121.6	475.5	557.3	1056.5	BS	475.5	1.10
	<i>d</i> -350- <i>L_s</i> -200	464.9	421.3	150.9	241.2	300.0	150.9	501.6	557.3	1244.7	BS	501.6	1.08
	<i>d</i> -350- <i>L_s</i> -250	480.4	585.2	179.1	241.2	300.0	179.1	526.6	557.3	1422.3	BS	526.6	1.10
	<i>d</i> -350- <i>L_s</i> -300	495.6	772.5	205.1	241.2	300.0	205.1	549.7	557.3	1508.1	BS	549.7	1.11
	<i>d</i> -450- <i>L_s</i> -150	555.3	280.9	121.6	241.2	300.0	121.6	651.5	557.3	1444.6	SS	557.3	1.00
	<i>d</i> -450- <i>L_s</i> -200	569.9	421.3	150.9	241.2	300.0	150.9	676.9	557.3	1631.0	SS	557.3	0.98
	<i>d</i> -450- <i>L_s</i> -250	573.5	585.2	179.1	241.2	300.0	179.1	701.3	557.3	1692.2	SS	557.3	0.97
G4	<i>d</i> -450- <i>L_s</i> -300	573.0	772.5	205.1	241.2	300.0	205.1	723.8	557.3	1743.9	SS	557.3	0.97
	<i>d</i> -250- <i>b_s</i> -150	351.6	438.9	160.7	188.2	300.0	160.7	341.1	557.3	658.9	BS	341.1	0.97
	<i>d</i> -250- <i>b_s</i> -200	374.5	585.2	179.1	241.2	300.0	179.1	358.0	557.3	673.6	BS	358.0	0.96
	<i>d</i> -250- <i>b_s</i> -250	407.6	731.5	197.4	294.2	300.0	197.4	374.9	557.3	949.5	BS	374.9	0.92
	<i>d</i> -350- <i>b_s</i> -150	473.8	438.9	160.7	188.2	300.0	160.7	510.3	557.3	1422.0	BS	510.3	1.08
	<i>d</i> -350- <i>b_s</i> -200	480.4	585.2	179.1	241.2	300.0	179.1	526.6	557.3	1422.3	BS	526.6	1.10
	<i>d</i> -350- <i>b_s</i> -250	488.0	731.5	197.4	294.2	300.0	197.4	542.9	557.3	1419.6	BS	542.9	1.11
	<i>d</i> -450- <i>b_s</i> -150	573.8	438.9	160.7	188.2	300.0	160.7	685.4	557.3	1702.9	SS	557.3	0.97
	<i>d</i> -450- <i>b_s</i> -200	573.5	585.2	179.1	241.2	300.0	179.1	701.3	557.3	1692.2	SS	557.3	0.97
G5	<i>d</i> -450- <i>b_s</i> -250	571.9	731.5	197.4	294.2	300.0	197.4	717.2	557.3	1679.8	SS	557.3	0.97
	<i>d</i> -250- <i>t_w</i> -4	313.7	585.2	126.3	226.6	300.0	126.3	309.5	557.3	673.6	BS	309.5	0.99
	<i>d</i> -250- <i>t_w</i> -8	374.5	585.2	179.1	241.2	300.0	179.1	358.0	557.3	673.6	BS	358.0	0.96
	<i>d</i> -250- <i>t_w</i> -16	472.7	585.2	284.6	270.3	300.0	270.3	441.7	557.3	673.6	BB	441.7	0.93
	<i>d</i> -350- <i>t_w</i> -4	447.0	585.2	126.3	226.6	300.0	126.3	479.7	557.3	1422.3	BS	479.7	1.07
	<i>d</i> -350- <i>t_w</i> -8	480.4	585.2	179.1	241.2	300.0	179.1	526.6	557.3	1422.3	BS	526.6	1.10
	<i>d</i> -350- <i>t_w</i> -16	505.7	585.2	284.6	270.3	300.0	270.3	607.7	557.3	1422.3	SS	557.3	1.10
	<i>d</i> -450- <i>t_w</i> -4	569.7	585.2	126.3	226.6	300.0	126.3	655.6	557.3	1692.2	SS	557.3	0.98
G6	<i>d</i> -450- <i>t_w</i> -8	573.5	585.2	179.1	241.2	300.0	179.1	701.3	557.3	1692.2	SS	557.3	0.97
	<i>d</i> -450- <i>t_w</i> -16	566.6	585.2	284.6	270.3	300.0	270.3	780.2	557.3	1692.2	SS	557.3	0.98
	<i>d</i> -250- <i>f_{cu}</i> -35	374.5	585.2	179.1	241.2	300.0	179.1	358.0	557.3	673.6	BS	358.0	0.96
	<i>d</i> -250- <i>f_{cu}</i> -50	434.9	836.0	209.3	241.2	321.4	209.3	460.1	561.4	766.5	BS	460.1	1.06
	<i>d</i> -250- <i>f_{cu}</i> -60	456.2	962.5	229.5	241.2	335.9	229.5	529.1	564.0	835.7	BS	529.1	1.16
	<i>d</i> -350- <i>f_{cu}</i> -35	480.4	585.2	179.1	241.2	300.0	179.1	526.6	557.3	1422.3	BS	526.6	1.10
	<i>d</i> -350- <i>f_{cu}</i> -50	533.9	836.0	209.3	241.2	321.4	209.3	694.7	561.4	1590.3	SS	561.4	1.05
	<i>d</i> -350- <i>f_{cu}</i> -60	552.2	962.5	229.5	241.2	335.9	229.5	808.4	564.0	1703.8	SS	564.0	1.02
G7	<i>d</i> -450- <i>f_{cu}</i> -35	573.5	585.2	179.1	241.2	300.0	179.1	701.3	557.3	1692.2	SS	557.3	0.97
	<i>d</i> -450- <i>f_{cu}</i> -50	586.8	836.0	209.3	241.2	321.4	209.3	889.0	561.4	1896.0	SS	561.4	0.96
	<i>d</i> -450- <i>f_{cu}</i> -60	601.0	962.5	229.5	241.2	335.9	229.5	1011.1	564.0	2032.0	SS	564.0	0.94
	<i>d</i> -250- <i>f_{ys}</i> -235	374.5	585.2	179.1	241.2	300.0	179.1	358.0	557.3	673.6	BS	358.0	0.96
	<i>d</i> -250- <i>f_{ys}</i> -355	429.0	585.2	234.4	364.3	300.0	234.4	408.8	557.3	673.6	BS	408.8	0.95
	<i>d</i> -250- <i>f_{ys}</i> -390	444.1	585.2	250.6	400.3	300.0	250.6	423.7	557.3	673.6	BS	423.7	0.95
	<i>d</i> -250- <i>f_{ys}</i> -420	453.1	585.2	264.4	431.0	300.0	264.4	436.4	557.3	673.6	BS	436.4	0.96
	<i>d</i> -350- <i>f_{ys}</i> -235	480.4	585.2	179.1	241.2	300.0	179.1	526.6	557.3	1422.3	BS	526.6	1.10
	<i>d</i> -350- <i>f_{ys}</i> -355	493.4	585.2	234.4	364.3	300.0	234.4	575.8	557.3	1422.3	SS	557.3	1.13
	<i>d</i> -350- <i>f_{ys}</i> -390	495.6	585.2	250.6	400.3	300.0	250.6	590.2	557.3	1422.3	SS	557.3	1.12
	<i>d</i> -350- <i>f_{ys}</i> -420	501.8	585.2	264.4	431.0	300.0	264.4	602.5	557.3	1422.3	SS	557.3	1.11
	<i>d</i> -450- <i>f_{ys}</i> -235	573.5	585.2	179.1	241.2	300.0	179.1	701.3	557.3	1692.2	SS	557.3	0.97
	<i>d</i> -450- <i>f_{ys}</i> -355	570.6	585.2	234.4	364.3	300.0	234.4	749.2	557.3	1692.2	SS	557.3	0.98
	<i>d</i> -450- <i>f_{ys}</i> -390	570.8	585.2	250.6	400.3	300.0	250.6	763.1	557.3	1692.2	SS	557.3	0.98
	<i>d</i> -450- <i>f_{ys}</i> -420	568.7	585.2	264.4	431.0	300.0	264.4	775.1	557.3	1692.2	SS	557.3	0.98

631 6. Conclusion

632 This study conducted cyclic loading tests on two embedded column base (ECB)

connections and two beam-strengthened embedded CFST column base (BSECB) connections. Based on the experimental results, a refined FE model was developed to simulate the behavior of BSECB connections. Parametric analyses were subsequently carried out to investigate the load-transfer mechanisms and failure modes. Finally, a analytical method was proposed to predict the ultimate strength of BSECB connections. The main conclusions are summarized as follows:

(1) Two typical failure modes were observed in cyclic tests of shallowly embedded column base connections: concrete foundation failure and CFST column failure. The incorporation of strengthening beams significantly mitigates pinching and sliding in the hysteretic response, while also reducing cracking in the concrete foundation.

(2) The addition of strengthening beams effectively enhances the seismic performance of shallowly embedded CFST column bases, including improvements in stiffness, ductility, and energy dissipation capacity. However, this beneficial effect diminishes with increasing embedment depth.

(3) When the failure is governed by the concrete foundation, increasing the embedment depth and applying a moderate axial load ratio (≤ 0.2) can improve the ultimate strength of BSECB connections. In contrast, when CFST column failure governs, the embedment depth has a negligible effect, and increasing axial load may adversely affect the load-carrying capacity.

(4) The contribution of strengthening beams to the ultimate strength is governed by four potential failure modes, among which strengthening beam shear failure is dominant in most cases. Accordingly, increasing the web thickness enhances the contribution of the strengthening beams. In addition, the concrete reaction acting on the beams does not increase indefinitely with beam length, but tends to plateau beyond a certain length.

(5) The proposed method shows good agreement with the FE results. The mean value of M_{pre}/M_{FEM} is 1.01, with a coefficient of variation of 0.07. The corresponding mean ratios for specimens governed by beam shear failure (BS) and CFST section strength (SS) are 1.03 and 1.00, respectively.

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